



Roberto Varas González

PhD in astrophysics



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<https://rvarasg.github.io/>

EDUCATION

2022 - 2026 **Ph.D. Astrophysics, University of Granada** (Granada, Spain)
Supervised by Pedro J. Amado (IAA) and Francisco J. Pozuelos (IAA)

Enhancing the radial velocity precision of high-resolution near-infrared spectrographs for detecting exoplanets around ultracool dwarfs, at the IAA-CSIC. Improvement of the CARMENES-NIR channel thermal stability. Development of a method for computing the projected rotational velocity of the CARMENES stellar sample. Estimation of rotational periods using photometric data (from TESS, OSN, ...). Participated in the SPECULOOS-3 b discovery. Design and implementation of opto-mechanical components for the OSN and CAHA observatories.

Sept 2020 - **M.Sc. Space Science and Technology, EHU/UPV** (Bilbao, Spain), *first class*
March 2022 Thesis supervised by Nicholas Devaney (University of Galway) and Alberto Ortega (EHU)
Thesis project for developing the optics of the GLIP, a polarimeter for the NRT (ORM, La Palma). Measurement of polarimetric parameters of astronomical sources in three color bands, without moving parts.

Sept 2016 - **B.Sc. Aerospace Engineering, UPM** (Madrid, Spain), *first class*
Sept 2020 Thesis supervised by Pablo Salgado (UPM) and Jeff Porter (UPM)
Thesis project at the E-USOC - Spanish User Support and Operations Centre. Analysis and CFM simulation of PCM systems under microgravity conditions. Lead to two peer-reviewed articles as first author.

RESEARCH EXPERIENCE AND EMPLOYEMENT

Nov - Dic **Research stay**, Institut für Astrophysik und Geophysik, University of
2023 Göttingen (Germany). Supervised by M. Zechmeister and A. Reiners

2022 - **PhD in Astrophysics**, Instituto de Astrofísica de Andalucía (Granada, Spain)
2026 Supervised by Pedro J. Amado and Francisco J. Pozuelos

2021 - **Instrumentation Engineer**, Instituto de Astrofísica de Andalucía
2026 Main projects: CARMENES (CAHA), TARSIS (CAHA), OSN, and ANDES (ELT)

July 2021 - **Internship**, Instituto de Astrofísica de Canarias, IAC (Tenerife, Spain)
Oct 2021 Development and testing of cryogenics in astrophysical detectors

June 2019 - **Project engineer**, Gravity-On global s.l. (Madrid, Spain) - Renewable energy
Nov 2021 start-up

AWARDS

2022 Academic excellence, elected by the masters committee (top of the class)
2020 Best project of CERN IdeaSquare Workshop "Design the Future" 2020
2020 Academic excellence, elected by the School of Aerospace engineering

RESEARCH INTERESTS

- Instrumentation/telescope development and maintenance, particularly spectrographs
- CAD and optical design of astrophysical instrumentation
- Spectroscopy and photometry of ultracool dwarfs and exoplanets around them
- Software and pipeline development for astrophysical purposes

SKILLS

Astronomical techniques

- CARMENES data reduction and analysis
- Photometry data reduction and analysis (TESS, Observatory of Sierra Nevada, ...)
- Spectroscopic data analysis (serval, dice, ...)

Instrumentation projects

- **ANDES** (ELT), optomechanics design, 2021-2025
- **TARSIS** (CAHA, Granada), optomechanics design, 2021-2026
- **CARMENES-PLUS** (CAHA), cryo-vacuum and software development, 2021-2026
- **OSN**, optomechanics design, 2021-2024
- **GLIP** (ORM, La Palma), optics design, 2021-2022
- **Observatory of Tenerife**, cryo-vacuum and optomechanics, 2021

Computer skills

- Languages: Python, Matlab, Latex, Fortran
- Astronomical software: serval, DS9, DACE, lightkurve, exostriker, prose
- Engineering software: Zemax OpticStudio, 3d-Experience, Catia, CREO, Ansys, comsol
- Operating systems: GNU/Linux, Windows

On-site observing

- **Observatorio de Sierra Nevada** (Granada, Spain). Visitor astronomer, 2025 (4 nights)
- **Gran Telescopio Canarias** (La Palma, Spain). Visitor astronomer, Megara, 2024 (3 nights)

Proposals and instruments

- **CARMENES** (CAHA 3.5m), Rocky worlds in the JWST Era: Precise densities to support atmospheric studies, PI: A. Psaridi, ~500h awarded (2025-present)
- **PANIC** (CAHA 2.2m), Probing SPECULOOS-3b with PANIC, PI: F. J. Pozuelos, 8h awarded (2025)
- **OSN** (Granada, Spain), *photometric rotational periods of ucDs*, PI: R. Varas, 118h awarded (2025)
- **Neo-Narval** (Pic du Midi, France), ZDI mapping of GJ-436, PI: D. Revilla, 17.4h awarded (2025)
- **CARMENES** (CAHA 3.5m), *SPECULOOS-3 radial velocity survey*, PI: P. J. Amado, 18h awarded (2024)
- **CARMENES** (CAHA 3.5m), *Giant planets orbiting M-dwarfs*, PI: F. J. Pozuelos, 4h awarded (2024)

Workshops and training

- SO training Introduction to Github, IAA, 2024 (Granada, Spain)
- SO IAA Writing & Communicating your science workshop, 2024 (Granada, Spain)
- English for Academic Purposes: a workshop for young researchers, IAA, 2022 (Granada)
- Infrared optomechanics systems and instrumentation, IAA, 2022 (Granada, Spain)
- Metrology arm: Pcdmiso CAD, IAA, 2022 (Granada, Spain)
- Dassault Systemes managing training, IAA, 2021 (Granada, Spain)

Memberships

- CARMENES consortium
- Sociedad Española de Astronomía (SEA) and European Astronomical Society (EAS)

Organizing roles

- LOC of the Spanish Astronomical Society 2024 meeting (Granada, Spain)
- SOC of An Intersectional Approach to EDI Strategies (SS51) at EAS 2025 (Cork, Ireland)

TALKS AND POSTERS

- 2026 **Selected talk**, XVII Scientific Meeting of the SEA: *A homogeneous catalogue of projected rotational velocities accounting for limb-darkening* (Tarragona, Spain)
- 2025 **Selected talk**, Europlanet Science Congress 2025: *Hunting exoplanets around ultracool dwarfs with near-infrared radial velocity spectrographs* (Helsinki, Finland)
- 2025 **Selected talk**, Brown Dwarfs Keep Their Cool: *Present and future of high-resolution spectroscopy of exoplanets around ultracool dwarfs* (La Gomera, Spain)
- 2025 **Selected talk**, Workshop EPRV 6: *Improving radial velocity precision with CARMENES-PLUS An upgrade of the near-infrared Spectrograph* (Porto, Portugal)
- 2025 **Selected talk**, European Astronomical Society meeting: *Hunting Exoplanets Around Ultracool Dwarfs with Near-Infrared Radial Velocity Spectrographs* (Cork, Ireland)
- 2025 **Selected poster**, European Astronomical Society: *Catalogue of Rotational Velocities of CARMENES M-dwarfs Including Limb Darkening* (Cork, Ireland)
- 2024 **Selected poster**, XVI Reunión Científica de la SEA: *Hunting exoplanets around ultracool dwarfs with RV NIR-spectrographs* (Granada, Spain)
- 2024 **Selected poster**, Exoplanets 5: *Hunting exoplanets around ultracool dwarfs with RV NIR-spectrographs* (Leiden, Netherlands)
- 2024 **Talk**, 19th CARMENES scientific meeting: *CARMENES-PLUS* (Tenerife, Spain)
- 2023 **Selected poster**, The Extreme Precision Radial Velocity (EPRV) 5 Conference: *MARCOT: A new concept of a large aperture telescope to feed CARMENES*
- 2023 **Talk**, 18th CARMENES scientific meeting: *NIR spectrograph's stability Update* (Heidelberg, Germany)
- 2023 **Seminar**, Instituto de Astrofísica de Andalucía: *CARMENES-PLUS: Towards an optimization of the NIR spectrograph cooling system* (Granada, Spain)
- 2022 **Selected poster**, IAA Severo Ochoa Meeting: *Addressing key astrophysical questions from Granada: Towards an optimization of the CARMENES-NIR channel: A continuous flow feeding of the cooling system* (Granada, Spain)
- 2022 **Talk**, 17th CARMENES scientific meeting: *Upgrades & performance of the CARMENES-NIR cooling system* (Almería, Spain)
- 2021 **Seminar**, Instituto de Astrofísica de Canarias: *Puesta en marcha del sistema termoeléctrico de enfriamiento para criogenia de detectores* (Tenerife, Spain)

SCIENTIFIC PUBLICATIONS

Refereed publications

- **R. Varas**, G. Morello, M. Zechmeister, *et al.*, “A catalogue of the projected rotational velocity of the CARMENES M-dwarf sample and the effect of limb darkening”, *Astronomy and Astrophysics*, vol. 709, A244, May 2026. DOI: [10.1051/0004-6361/202659177](https://doi.org/10.1051/0004-6361/202659177)
- **R. Varas**, R. Calvo Ortega, P. J. Amado, *et al.*, “Improving radial velocity precision with CARMENES-PLUS: An upgrade of the near-infrared spectrograph cooling system”, *Experimental Astronomy*, vol. 60, September 2025. DOI: [10.1007/s10686-025-10020-0](https://doi.org/10.1007/s10686-025-10020-0)
- M. Gillon *et al.* (including **R. Varas**), “Detection of an Earth-sized exoplanet orbiting the nearby ultracool dwarf star SPECULOOS 3”, *Nature Astronomy*, vol. 8, 865-878, May 2024. DOI: [10.1038/s41550-024-02271-2](https://doi.org/10.1038/s41550-024-02271-2)

- **R. Varas**, Ú. Martínez, Karl Olfe, *et al.*, “Effects of Thermocapillary and Natural Convection During the Melting of PCMs with a Liquid Bridge Geometry”, *Microgravity Science and Technology*, vol. 35, A17, March 2023. DOI: [10.1007/s12217-023-10040-5](https://doi.org/10.1007/s12217-023-10040-5)
- **R. Varas**, P. Salgado Sánchez, J. Porter, *et al.*, “Thermocapillary effects during the melting in microgravity of phase change materials with a liquid bridge geometry”, *International Journal of Heat and Mass Transfer*, vol. 178, 121586, October 2021. DOI: [10.1016/j.ijheatmasstransfer.2021.121586](https://doi.org/10.1016/j.ijheatmasstransfer.2021.121586)
- H. L. Ruh *et al.* (including **R. Varas**), “The CARMENES search for exoplanets around M dwarfs - The impact of rotation and magnetic fields on the radial velocity jitter in cool stars”, *A&A*, vol. 692, A138, December 2024. DOI: [10.1051/0004-6361/202450836](https://doi.org/10.1051/0004-6361/202450836)

Proceedings

- **Varas, R.**, Amado, P. J., and Pozuelos, F. J.: Hunting exoplanets around ultracool dwarfs with near-infrared radial velocity spectrographs, EPSC-DPS 2025, Helsinki, Finland, 7–12 Sep 2025, EPSC-DPS2025-1790, <https://doi.org/10.5194/epsc-dps2025-1790>, 2025.
- **R. Varas**, P. J. Amado, R. Calvo-Ortega, and M. Centenera, “Hunting exoplanets around latest M dwarfs with RV NIR-spectrographs”, *XVI Scientific Meeting of the Spanish Astronomical Society*, July 2024. DOI: [10.5281/zenodo.14012840](https://doi.org/10.5281/zenodo.14012840)
- A. Marconi *et al.* (including **R. Varas**), “ANDES, the high resolution spectrograph for the ELT: science case, baseline design and path to construction”, *Ground-based and Airborne Instrumentation for Astronomy IX (SPIE)*, August 2022. DOI: [10.1117/12.2628689](https://doi.org/10.1117/12.2628689)
- E. G. P. O'Connor, É. J. Harvey, I. A. Steele, N. Devaney, & **R. Varas**, “Galway Liverpool Imaging Polarimeter – GLIP: design and prototype status”, *Ground-based and Airborne Instrumentation for Astronomy IX (SPIE)*, August 2022. DOI: [10.1117/12.2629283](https://doi.org/10.1117/12.2629283)

OUTREACH AND SOCIAL INVOLVEMENT

2024 - present	Member of the Diversity, Equity and Inclusion committee at the IAA
2024 - present	Member of the Sustainability committee at the IAA
2024 - present	Member of the editorial board of the outreach journal of the IAA
2024	Volunteer in PIISA project (Granada, Spain)
2023 - 2024	Volunteer at Espacio 3 (Granada, Spain)
2023	Volunteer at the 100 x Ciencia.7 conference (Granada, Spain)
2023	Talk at Espacio 3 (Granada, Spain)
2023	Talk at European Researcher Night (Granada, Spain)
2023	Volunteer at the European Researcher Night (Granada, Spain)
2023	Talk at Pint of Science (Granada, Spain)
2022	Volunteer at the IAA Severo Ochoa Meeting (Granada, Spain)

REFERENCES

- Dr. David Pérez Medialdea, Instituto de Astrofísica de Andalucía, davidp@iaa.es
- Dr. Pedro José Amado, Instituto de Astrofísica de Andalucía-CSIC, pja@iaa.es
- Dr. Francisco José Pozuelos, Instituto de Astrofísica de Andalucía-CSIC, pozuelos@iaa.es
- Dr. Mathias Zechmeister, Institut für Astrophysik Göttingen, zechmeister@astro.physik.uni-goettingen.de